# Reiner Ethan Umila

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## PROFESSIONAL SUMMARY

Software Engineering student at McMaster University currently completing an 8-month Software Quality Assurance co-op with the Government of Ontario, performing manual and automated QA testing on OSAP's student- and administrator-facing web applications, tracing root-cause defects following a platform migration to Azure, overhauling long-neglected test documentation, and building self-directed Python Playwright automation to accelerate manual test execution. Hands-on experience also spans full-stack development, data engineering, and embedded systems: designed and deployed live a full-stack Notion-style productivity app (Next.js, TypeScript, Supabase) with a dynamic-schema database and a layered Row Level Security and GRANT permission model, and a separate full-stack weather application with a FastAPI Python back-end, pandas ETL pipeline, Pydantic validation models, SQLite caching, and a fully styled React front-end with automatic dark mode and card-based hourly forecast display. Additional experience in algorithmic Python development with NumPy, Streamlit dashboard design, REST API integration, SQL, object-oriented programming, UML design, automated testing with Selenium, AI-assisted development workflows using Claude, and technical documentation. Hardware background includes ESP32 firmware in C++, custom PCB design in Autodesk Fusion, $I^2C$/SPI/MQTT integration, and Node-RED IoT dashboards from a formal capstone project.

## EDUCATION

**Bachelor of Technology, Software Engineering Technology (Co-op)**
McMaster University — Expected December 2027

**Ontario College Advanced Diploma, Computer Engineering Technology**
Seneca Polytechnic — 2024

- 3.8 GPA, Honours

## PROJECTS

**Noctus (Notion-Style Productivity App)** | Next.js, TypeScript, Tailwind CSS, shadcn/ui, Supabase, Vercel
*Live Personal Project* — *2026 – Present*

- Designed, built, and deployed a full-stack, Notion-style productivity app live on Vercel: a dynamic-schema Postgres database (via Supabase) supporting five user-defined column types (text, number, date, boolean, select), with full CRUD across boards, properties, and entries, and type enforcement handled at the database layer rather than the application layer.
- Implemented a two-layer permission model combining Row Level Security policies, some tracing data ownership through two table joins, with explicit Postgres GRANT statements as a separate, independently enforced access check, and built the dynamic table interface on TanStack Table with type-aware column logic driven entirely by each board's user-defined schema.
- Independently picked up several technologies for the first time to ship this, including a managed cloud database, OAuth, Row Level Security, and the Next.js App Router, going from a broken build to a working, deployed product within days once the core blocker was identified and resolved.

**Weather Application (Full-Stack)** | Python, FastAPI, React, JavaScript, pandas, Pydantic, SQLite, HTML/CSS
*Live Personal Project* — *December 2025 – Present*

- Designed, built, and deployed live (Netlify, Render) a complete client/server weather application (v1.0): a FastAPI Python back-end with two RESTful API endpoints, CORS middleware, structured exception handling, and Pydantic data models (CurrentWeather, HourlyWeather, WeatherResponse) enforcing validated response contracts at the API boundary.
- Architected two back-end data retrieval paths: a city-name endpoint geocoding user input via the Nominatim API before querying Open-Meteo across 8 meteorological variables, and a coordinates endpoint accepting raw browser geolocation values with reverse geocoding at town/borough granularity.
- Applied pandas to run an ETL-style data pipeline on raw hourly forecast data: constructing timezone-aware datetime indices from UTC timestamps via tz_convert(), filtering a 168-hour raw dataset to a 24-hour rolling forward window, rounding and type-casting all fields, and serialising the result for front-end consumption; implemented SQLite-backed caching with 1-hour expiry and retry logic to reduce redundant API calls.
- Built a fully styled React front-end with a CSS custom property design system supporting automatic light and dark mode, card-based hourly forecast display, a hand-rolled SVG precipitation chart, dual-mode location input (browser geolocation on mount plus manual city search form), full async state management via React hooks, and conditional rendering across loading, error, and data states.

**BlockBlast! Puzzle Solver** | Python, NumPy, Streamlit

*Live Personal Project* *February 2026 – Present*

- Designed a modular, object-oriented codebase across four files with clean separation of concerns: solver orchestration, board mechanics, piece catalog, and GUI layer, each independently testable and replaceable.
- Built a recursive, permutation-aware search algorithm exploring every ordering of the three available pieces, since placement order affects which lines clear and what space opens up next, applying a depth-first search at each ordering across an 8x8 NumPy board representation to find the highest-scoring complete placement sequence.
- Implemented a piece-recognition system that crops user input to its bounding box and validates it against a full catalogue of named NumPy array shapes before the solver runs, and built a fully interactive Streamlit GUI, repurposing its data-display widgets into a clickable board editor with colour-coded, step-by-step solution playback and multi-turn session state continuity, deployed live on Streamlit Community Cloud.
- Developed a test suite validating solver output against known board states and applied a continuous test-and-refine cycle to surface and fix edge case failures at each development stage.

**Human Benchmark Game Device (Capstone)** | C++, ESP32-S3, Node-RED, Autodesk Fusion, PlatformIO
*Academic Project, Seneca Polytechnic* *November 2024*

- Collaborated with a partner across the full project lifecycle within a 2.5-month timeline: requirements, system design, C++ firmware on the ESP32-S3, two custom PCBs designed in Autodesk Fusion, hardware integration, testing, and formal delivery to an evaluation panel.
- Diagnosed and resolved multiple hardware integration issues by tracing root causes through datasheet analysis and targeted testing, resolving $I^2C$, SPI, and GPIO conflicts; integrated a Node-RED IoT dashboard for real-time session performance tracking and data logging via MQTT.

## WORK EXPERIENCE

**Software Quality Assurance Intern (Co-op)**, Ministry of Public and Business Service Delivery and Procurement, Toronto, ON May 2026 – December 2026

- Perform manual and automated QA testing on OSAP's Student-Facing (SFA) and Administrator-Facing (AFA) web applications for the Government of Ontario, covering Full-Time, Part-Time, Microcredential, and OLSG/OLSG-ME financial aid application types.
- Log and track defects through HP's ALM within a structured, lead-driven QA lifecycle, coordinating fix verification and retesting with development teams across successive builds.
- Independently developed Python Playwright automation scripts to accelerate repetitive manual test cases, such as student profile creation, supplemented by direct hands-on exposure to the team's legacy automation tooling, UFT with VBScript, even as a manual tester.
- Traced software defects in older academic-year application logic back to their root cause following the platform's migration to Azure, surfacing data quality issues through rigorous regression testing rather than routine execution alone.
- Rewrote and streamlined test case documentation left unrevised for up to five years, correcting unclear or inaccurate pre-conditions and execution steps to improve usability for future co-op cohorts.

**Warehouse Attendant**, University of Toronto Press, Toronto, ON 2022 – 2026

- Identified the root cause of a persistent inventory backlog in the warehouse management system (AS400), developed a solution independently, and executed both the physical reorganisation and system updates, reducing the required cleanup cycle from weekly to monthly.
- Regularly called upon by colleagues to diagnose and resolve technical issues with Windows workstations, printers, and AS400 warehouse software, drawing on self-acquired knowledge to keep operations running.

## ADDITIONAL SKILLS

**Languages:** Python, JavaScript, TypeScript, C++, C, HTML/CSS, SQL

**Frameworks:** FastAPI, React, Next.js, Pydantic, Streamlit, pandas, NumPy, Supabase, Tailwind CSS, shadcn/ui

**Testing:** Manual and automated QA testing, Playwright (Python, self-directed automation), Selenium (automated web testing), UFT/VBScript (legacy exposure), HP ALM defect tracking, root-cause analysis, functional testing, system testing, UML (use case, class, sequence, state diagrams)

**Networking & IT:** DNS, HTTP/HTTPS, TCP/IP, Kali Linux, Security Onion, Wireshark, Cisco Packet Tracer, Active Directory

**Concepts:** OOP, REST APIs, full SDLC, Agile, Waterfall, Iterative, client/server architecture, data modelling, relational database design, ETL/ELT pipelines, QA methodology, STLC, OAuth, Row Level Security

**Tools:** Git, VSCode, PlatformIO, Node-RED, Autodesk Fusion, Vercel, Netlify, Render, Supabase, Claude (AI-assisted development), Microsoft Office (Word, Excel, Outlook, PowerPoint), Microsoft Teams, SharePoint, SAP (basic), AS400 (basic)